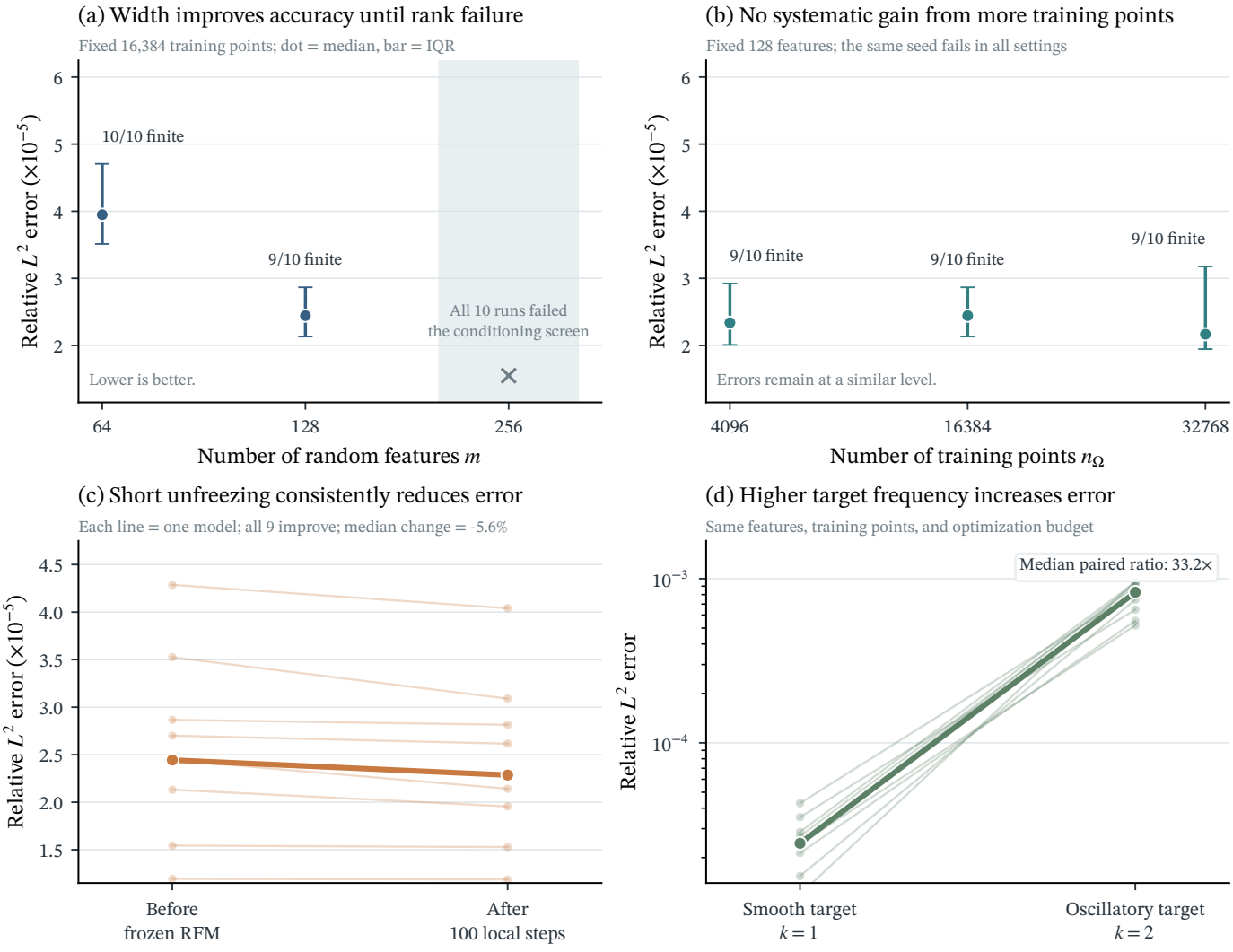
(a) Width improves accuracy until rank failure
Fixed 16,384 training points; dot = median, bar = IQR
Relative L^2 error ($\times 10^{-5}$)
10/10 finite
9/10 finite
All 10 runs failed
the conditioning screen
Lower is better.
64
128
256
Number of random features m
(b) No systematic gain from more training points
Fixed 128 features; the same seed fails in all settings
9/10 finite
9/10 finite
9/10 finite
Errors remain at a similar level.
4096
16384
32768
Number of training points n_Ω
(c) Short unfreezing consistently reduces error
Each line = one model; all 9 improve; median change = -5.6%
Before
frozen RFM
After
100 local steps
(d) Higher target frequency increases error
Same features, training points, and optimization budget
Median paired ratio: 33.2×
Smooth target
$k = 1$
Oscillatory target
$k = 2$